

Chapter 7: National Income Accounting

Meaning of National Income

National income refers to the total income earned by all individuals of a country in one year.

Main Concepts of National Income

1. **GDP (Gross Domestic Product):**

Market value of all final goods and services produced within the country in a year.

$$\text{GDP} = P_1Q_1 + P_2Q_2 + \dots P_nQ_n$$

2. **GNP (Gross National Product):**

GDP plus net factor income from abroad.

$$\text{GNP} = \text{GDP} + \text{NFIA}$$

Where, NFIA = Net Factor income from abroad

3. **NNP (Net National Product):**

GNP minus depreciation (wear and tear of capital).

$$\text{NNP} = \text{GNP} - \text{Depreciation}$$

4. **NI (National Income):**

$$\text{NNP at factor cost} = \text{NNP} - \text{indirect taxes} + \text{subsidies.}$$

Or,

$$\text{NI} = \text{Sum of factor rewards} + \text{NFIA}$$

Or,

$$(W + R + I + P) + \text{NFIA}$$

Where,

NI = National Income

W = Wages and salaries

R = Rent

I = Interest

P = Profit

NFIA = Net factor income from abroad

Calculation of NI

Product Method

$$\begin{aligned}\text{GDP} &= \text{P}_1\text{Q}_1 + \text{P}_1\text{Q}_1 + \dots \text{P}_n\text{Q}_n \\ \text{GNP} &= \text{GDP} + \text{NFIA} \\ \text{NNP at FC} &= \text{NNP} - \text{Net indirect taxes} \\ \text{NI} &= \text{NNP at FC}\end{aligned}$$

Income Method

$$\begin{aligned}\text{GDP} &= (\text{W} + \text{R} + \text{I} + \text{P}) + \text{Depreciation} + \text{Net indirect taxes} \\ \text{GNP} &= \text{GDP} + \text{NFIA} \\ \text{NNP} &= \text{GNP} - \text{Depreciation} \\ \text{NNP at FC} &= \text{NNP} - \text{Net indirect taxes} \\ \text{NI} &= \text{NNP at FC}\end{aligned}$$

Expenditure Method

$$\begin{aligned}\text{GDP} &= \text{C} + \text{I} + \text{G} + (\text{X} - \text{M}) \\ \text{GNP} &= \text{GDP} + \text{NFIA} \\ \text{NNP} &= \text{GNP} - \text{Depreciation} \\ \text{NNP at FC} &= \text{NNP} - \text{Net indirect taxes} \\ \text{NI} &= \text{NNP at FC}\end{aligned}$$

5. **PI (Personal Income):**

Total income received by individuals before paying direct taxes.

$$\text{PI} = \text{NI} - \text{Undistributed corporate profit} - \text{Corporate income tax} - \text{Social security contribution} + \text{Transfer payment}$$

6. **DI (Disposable Income):**

Income left after paying direct taxes.

$$(\text{DI} = \text{PI} - \text{direct taxes})$$

7. **PCI (Per Capita Income):**

Average income per person in a year.

$$(\text{PCI} = \text{NI} \div \text{total population})$$

Nominal GDP, Real GDP, and GDP Deflator

1. **Nominal GDP:**

GDP measured at current year prices.

2. **Real GDP:**

GDP measured at base year prices (shows actual growth).

3. **GDP Deflator:**

Measures price changes (inflation) between base year and current year.

$$\text{GDP Deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$

Methods of Measuring National Income

1. **Product Method:**

National income = total value of final goods & services.

- *Final Product Method:* Value of all final goods and services.
- *Value Added Method:* Sum of value added at each production stage.

Gross Value added = Value of output – Cost of intermediate goods

2. **Income Method:**

National income = sum of all incomes earned.

Includes:

- Wages and salaries
- Rent
- Interest
- Profit
- Net indirect taxes
- Net income from abroad
- Depreciation

3. **Expenditure Method:**

National income = total expenditure on final goods and services.

Components:

- Personal consumption
- Government expenditure
- Private investment
- Net exports (export–import)

- Net indirect taxes
 - Net income from abroad
 - Depreciation
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Difficulties in Measuring National Income in Nepal

1. **Double Counting:** Same goods counted more than once.
2. **Depreciation:** Hard to estimate capital wear and tear.
3. **Price Changes:** Inflation affects comparison.
4. **Unreliable Data:** Lack of accurate records.
5. **Illegal Income:** Hidden or unreported earnings.
6. **Choice of Method:** Difficult to select suitable method.
7. **Non-market Activities:** Household and unpaid work not counted.
8. **Unreported Income:** People hide income to avoid taxes.
9. **Intermediate Goods:** Risk of counting these twice.
10. **Non-monetized Sector:** Many rural transactions are barter-based.
11. **Illiteracy & Ignorance:** People can't give correct information.
12. **Lack of Occupational Specialization:** Difficult to classify income sources properly.